

Aryaman Sarcar

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EDUCATION

Northeastern University, Houry College of Computer Science Boston, MA
B.S Computer Science (Artificial Intelligence) w/ Math Minor May 2028
Coursework: Object-Oriented Design, Discrete Maths, Differential Equations & Linear Algebra, Foundations Data Science, Probability & Statistics, Calculus 1 & 2, Algorithms & Data
Clubs: Northeastern Electric Racing Club, AI Club

TECHNICAL SKILLS

Languages: Python, Java, C++, R, JavaScript, TypeScript, SQL, MATLAB
Framework & Libraries: Flask, FastAPI, Quantlib, LangChain, LangGraph, PyTorch, Pandas, NumPy, Scikit, React, Tableau
Databases: PostgreSQL, MySQL
Developer Tools: AWS, Git, GCP, Linux, CI/CD, Git, REST APIs, Google Vertex AI, OpenAI, Bloomberg API

WORK EXPERIENCE

IBM San Jose, CA
Forward Deployed Engineer Co-op Sep 2026 – Present (Incoming)

Yellow Wood Partners LLC Boston, MA
AI Summer Analyst June 2026 – Aug 2026 (Incoming)

Navikenz Inc Princeton, NJ
AI Engineering Co-op Sep 2025 – March 2026

- Designed and tested AI agents for internal company platforms, building **LangChain**-based tool-execution workflows to automate structured data retrieval, analysis, and reporting tasks.
- Developed an agentic portfolio optimization and risk analysis system in Python, implementing **covariance modeling** and constrained **mean-variance** optimization using **cvxpy** and **PyPortfolioOpt**.
- Collaborated with the Chief Architect of Data & AI to refine agent orchestration patterns, improve system reliability, and formalize reproducible research data pipelines.

NEU Electric Racing Club Boston, MA
Software Team Member Sep 2024 - Dec 2024

- Contributed to the development of FinishLine, a full-stack web app used for project tracking project progress.
- Implemented frontend features using **Typescript** and collaborated on backend **API** development.
- Collaborated in a cross-functional team using Git for version control.

Vultus Inc South Plainfield, NJ
Software Intern May 2023 - Aug 2023

- Translated client requirements into a modern stock-tracking application prototype, independently designing UI wireframes that guided stakeholder discussions.
- Developed a functional proof-of-concept application with minimal supervision, collaborating with senior engineers in an **Agile** environment.
- Demonstrated critical thinking, programming, and adaptability to learn new tools and concepts.

PROJECTS

Quantitative Portfolio Management System – Professional Project October 2025 - Present

- Built a full-stack AI-driven agentic portfolio optimization system integrating Python, **Streamlit**, **FastAPI**, enabling live market data visualization via **yfinance** and **TradingView API**.
- Engineered multi-agent function (Data, Risk, Optimizer, Backtester, and Reporter Agents) to automate data retrieval, risk analysis, and **mean-variance** optimization using **PyPortfolioOpt** and **cvxpy**.
- Designed real-time analytics dashboards to simulate a full report on optimized portfolio with an interactive dashboard.

Market Shock Simulator – Personal Project March 2026 - Present

- Developed an interactive financial stress-testing dashboard using **React** and a custom **D3.js force simulation engine**, mapping portfolio asset weight to particle size and sector classifications to gravitational force centers.
- Integrated real-time market data from the **Alpha Vantage API** and designed a **SQLite** database to securely store user accounts, trade history, and portfolio holdings.
- Implemented a basic algorithmic trading strategy (moving average crossover) and added data visualization with **Chart.js**, enabling users to view price trends and back test simple strategies.

ADDITIONAL INFORMATION

Interests: Soccer, Tennis, Wrestling, Brazilian Jui-Jitsu, Lifting, Cooking, Football